jcm-strat: Phases 0 to 4

Approach A, a stripped-down stratosphere model on JCM / Dinosaur semi-Lagrangian. Status record as of 3 September 2026. Repository reflective-org/jcm-strat, working copy /data/JCM_stripped/jcm-strat.

The idea in one paragraph

Start from the full JCM climate model, remove every piece of tropospheric physics, replace what the troposphere needs with ERA5 nudging, then add passive tracers to measure stratospheric transport. Each phase adds or removes exactly one thing, so any change in behaviour has one cause. The target for Part 1 is a model that runs years of stratospheric tracer transport per hour on one H100, conserves tracer mass, stays non-negative, and produces an age-of-air pattern comparable to CLaMS. All runs use GPU 0 of the shared 8x H100 node, one at a time, on the T63L95 grid (1.9 degrees, 95 hybrid levels, top near 0.01 hPa) with a 12 minute time step.

Phase overview

Phase	What changed	Run length	Speed (sim. days / hour, stepping)	Status
0	Nothing. Stock JCM, full ECHAM physics	10 days	52	merged (PR 16)
1	Removed all 12 ECHAM terms; kept Held-Suarez only (dry)	1 year	5 787	merged (PR 21)
2	Added ERA5 nudging of u, v, T below 150 hPa; ERA5 initial state	1 year (2005)	5 455	merged (PR 22)
3	Added four passive tracers (aoa, unity, sai, e90)	1 year (2005)	4 458	merged (PR 24)
4	Nothing physical; run 2005-2009 as five chained one-year segments; age of air vs CLaMS and WACCM	5 years	4 315 (4 460-4 480 per segment)	done, PR open

The Phase-1 number (from Phase 4)

The stripped model runs one simulated year in 10.4 minutes end to end on one H100 (about 4 400 simulated days per hour stepping, 1 800 end to end), so 30 years cost about 5 GPU-hours; full ECHAM physics on the same grid takes 7.8 hours per year. Passive-tracer mass closes to 0.02 percent over five years for a uniform tracer and to 1 percent against an analytic source, with exact non-negativity and no polar pull-up. The age-of-air pattern is right (tropical pipe, poleward ageing) but the tropics are 1.5 years too old at 20 km and there is no polar-night jet: with Held-Suarez standing in for radiation the Brewer-Dobson circulation is too weak. The speed question is answered; the fidelity question now hinges on the stratospheric forcing (issues 2 and 5).

Phase 0: reproduce the stock model

Nothing of ours. Stock JCM with the full ECHAM physics package (RRTMGP radiation, Tiedtke convection, clouds, TKE diffusion, surface scheme, Hines and orographic gravity-wave drag) on the T63L95 grid for 10 days. Purpose: prove the environment is sound and set the reference speed.

- Environment: uv-managed Python 3.11, JAX 0.10.2 with CUDA 12, dinosaur 1.4.0 from the pinned semi-lagrangian branch (bd99e39b), JCM 2.1.0b0 from the pinned dev branch (849893b). Everything, including caches and runs, lives under /data because the root disk is 97 percent full.
- Result: 51.6 simulated days per hour at a 12 minute step and 51.9 at 15 minutes. The published A100 reference for this configuration is 52. Wall time per simulated day is independent of the step because radiation, firing every 2 hours, dominates the cost. The speed lever is therefore removing physics, not lengthening the step.
- One real bug found and fixed: JAX silently ran on CPU until the containerised libcuda path was added to the loader path. The environment check now asserts a CUDA device.

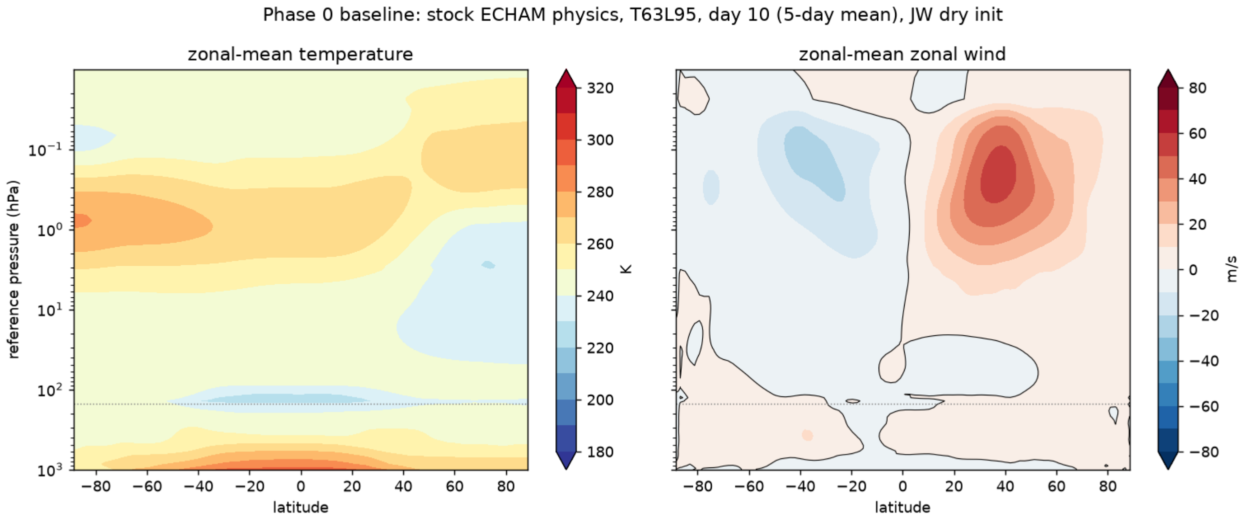

Figure 1. Phase 0 sanity plot: zonal-mean temperature and zonal wind at day 10 of the stock model from the dry Jablonowski-Williamson state. A spin-up, not a climatology.

Phase 1: strip the physics (dry Held-Suarez)

Removed: all 12 ECHAM terms. Convection, cloud fraction, cloud microphysics, boundary-layer diffusion, surface exchange, RRTMGP radiation, aerosol, chemistry, and both gravity-wave schemes. Kept: Held-Suarez (1994) only, that is Newtonian relaxation of temperature toward a prescribed profile plus Rayleigh friction near the surface. The model is dry: it carries no water, has no moisture physics, and starts from a zero-humidity Jablonowski-Williamson state.

Held-Suarez acts at every level. Below about the tropopause the equilibrium temperature has real structure (warm tropics, cold poles). Above it the target is clamped to a flat 200 K at every latitude and the relaxation time is 40 days. The stratosphere is therefore pulled toward an isothermal state with no seasonal cycle and no polar-night jet. A separate upper sponge relaxes the top ten levels toward 250 K as a numerical lid.

- One year, no NaNs in any of 13 chunks, surface pressure drift +0.005 hPa, top-level temperature flat.
- Speed: 5 787 simulated days per hour stepping, about 110 times Phase 0. A simulated year takes 9 minutes end to end including compile and output.
- Stratosphere is 21 K RMS colder than ERA5 between 7 and 70 hPa and has essentially no zonal wind. This is the known limitation of Held-Suarez, not a bug; fixes are filed as issues 2 (RRTMGP) and 5 (Polvani-Kushner).
- The planned gravity-wave-drag variant could not be composed with Held-Suarez on JCM's 3-D physics path and was dropped (issue 20). Phase 3 later removed that blocker.

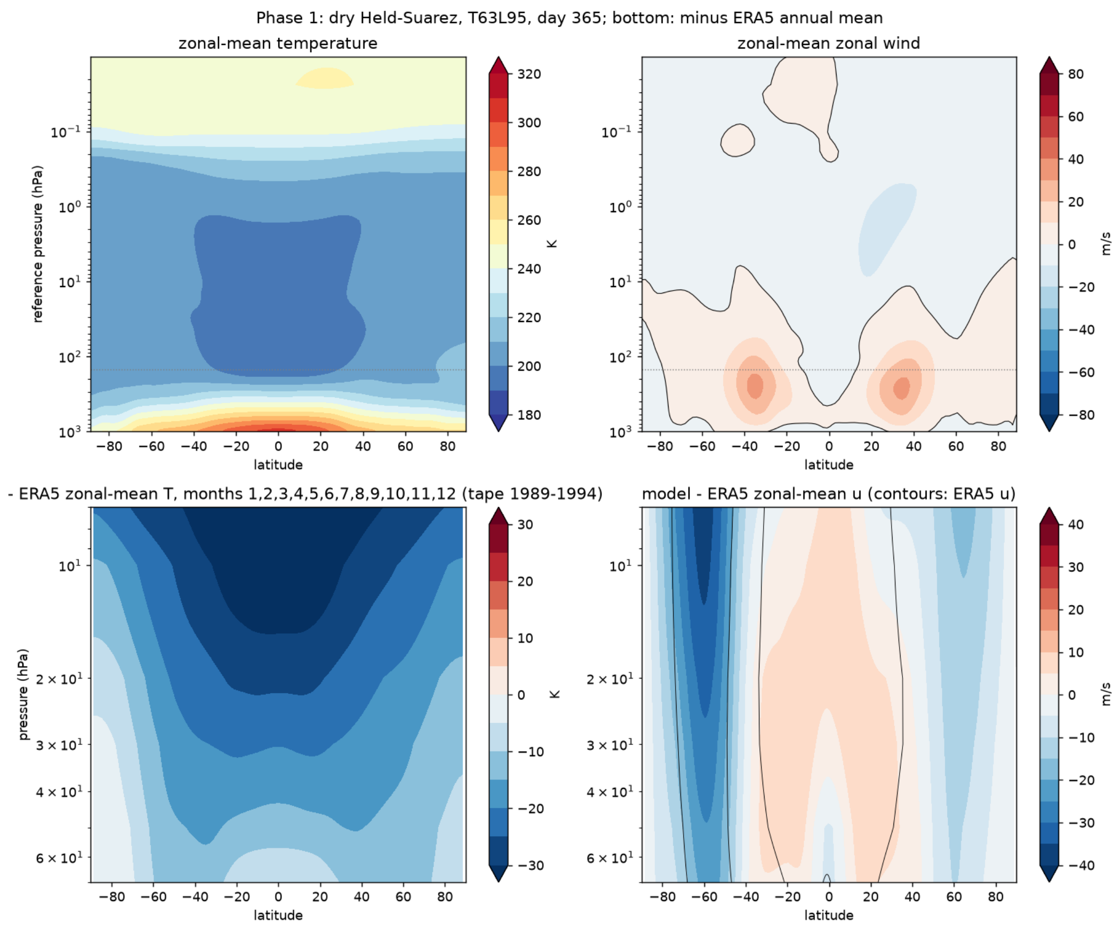

Figure 2. Phase 1 after one year. Top: zonal-mean temperature and zonal wind. Two symmetric subtropical jets near 200 hPa, an isothermal 200 K stratosphere, no polar-night jet. Bottom: model minus the ERA5 1989-1994 annual-mean zonal-mean tape over 7-70 hPa, the stratospheric cold bias and the missing stratospheric winds.

Phase 2: nudge the troposphere to ERA5 (specified dynamics)

Added: the initial state from ERA5 on 1 January 2005, and Newtonian nudging of the zonal wind, meridional wind and temperature toward 6-hourly ERA5 with a 6 hour relaxation time. Nothing else changed. The nudging is masked off in the lowest two model levels (boundary layer free) and at every level with reference pressure below 150 hPa, so the whole stratosphere evolves freely. Surface pressure, humidity and tracers are never nudged.

- One year (2005), stable, surface pressure drift -0.02 hPa, speed within 6 percent of Phase 1.
- The troposphere now carries real 2005 meteorology: global kinetic energy has a seasonal cycle, high in boreal winter and lowest around day 150, which the seasonless Phase 1 could not produce. Jets are asymmetric between hemispheres in late December.
- No visible discontinuity at the 150 hPa cutoff in temperature or wind.
- Stratosphere above 150 hPa is unchanged from Phase 1 by construction: still Held-Suarez, no polar-night jet. Planetary waves from the nudged troposphere can propagate up, but there is no radiatively driven vortex for them to act on.
- Not yet measured: RMSE against ERA5 at 500 hPa, the tropopause position, and the January 2006 SSW. The diagnostic scripts are issue 17.

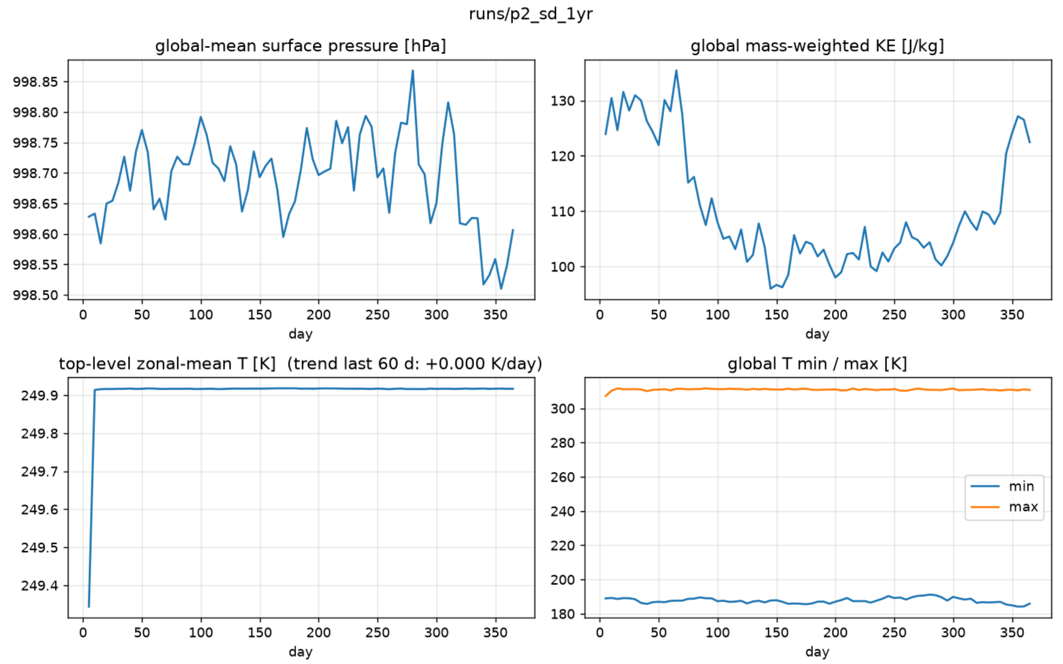

Figure 3. Phase 2 stability summary over 2005: global-mean surface pressure, global kinetic energy, top-level temperature, global temperature extremes. The seasonal cycle in kinetic energy is the signature of the nudged troposphere.

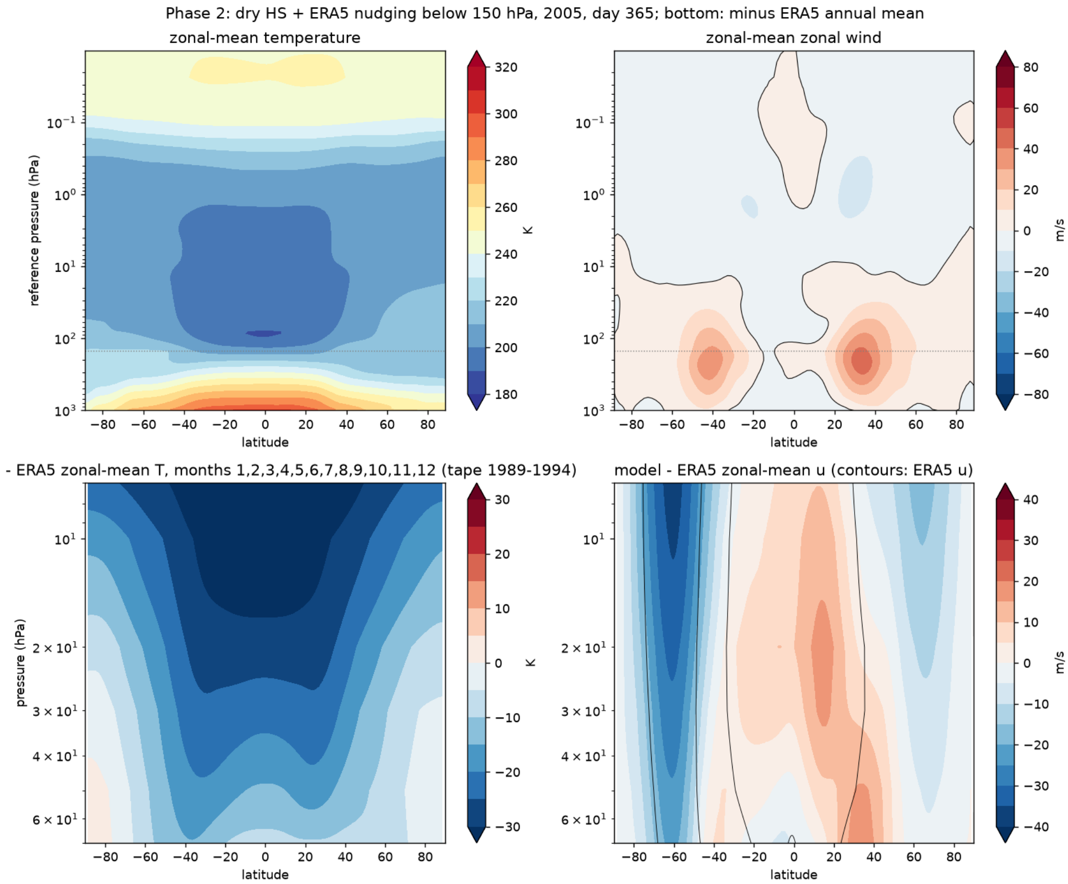

Figure 4. Phase 2 at day 365 (late December 2005). Top: hemispherically asymmetric jets, no kink at 150 hPa (dotted line). Bottom: the stratospheric bias against ERA5 is the same as in Phase 1, as it must be.

Phase 3: passive tracers

Added: one physics term carrying four passive tracers, advected by the semi-Lagrangian dycore as grid-point (nodal) fields with JCM's global proportional mass fixer on. Tracers ride on the winds and never feed back on the dynamics, so the meteorology is identical to Phase 2. Each tracer isolates a different property of the transport scheme:

Tracer	Definition	Question it answers	Result after 1 year
unity	1 everywhere, no sources or sinks	Does the scheme move air without inventing or destroying it? Any deviation is numerical error.	Worst cell off by 2.6e-4. Pass.
sai	Constant source in 15S-15N, 25-55 hPa (about 20-25 km); no sink	Is mass conserved with a known analytic answer (burden = source x box mass x time)? Also: minimum exactly zero, no polar pull-up at the top.	Tracks the analytic line to 1.4 percent; min 0.0; polar top-level value 0.1 percent of the global column. Pass.
aoa	Clock: +1 day per day everywhere, reset to 0 where p > 700 hPa	How long has this air been in the stratosphere? The mean age of air, the primary transport metric.	0.97-0.99 yr above 100 hPa: correct but uninformative after one year. Needs Phase 4.
e90	100 in the two lowest layers, 90-day e-folding sink (Prather et al. 2011)	Where is the tropopause? The 90 contour is a standard dynamical tropopause marker.	Does not work here: the dry model has no convection to stir the troposphere, so the 90 contour sits at 970 hPa. Issue 23.

The genuine find. The first attempt lost 56 percent of the clock over the year. JCM's global mass fixer restores each tracer's global mass after every step. At the sharp 700 hPa reset edge the limiter creates a little mass each step, and the fixer removed it by rescaling the whole field, stratosphere included. A clock is not a conserved quantity and must not be mass-fixed. It was invisible in every conservation diagnostic (unity, sai and e90 were all fine). Fixed with a config key that exempts aoa from the fixer, verified by a 5-day run and a unit test.

- Speed: 4 458 simulated days per hour stepping; four extra tracers cost about 1 ms on a 6 ms step. End to end a simulated year takes 10.5 minutes.
- Held-Suarez had to be re-implemented as a per-column variant with identical forcing, because JCM's 3-D physics path cannot add tracer tendencies across terms. This also unblocks gravity-wave drag (issue 20).

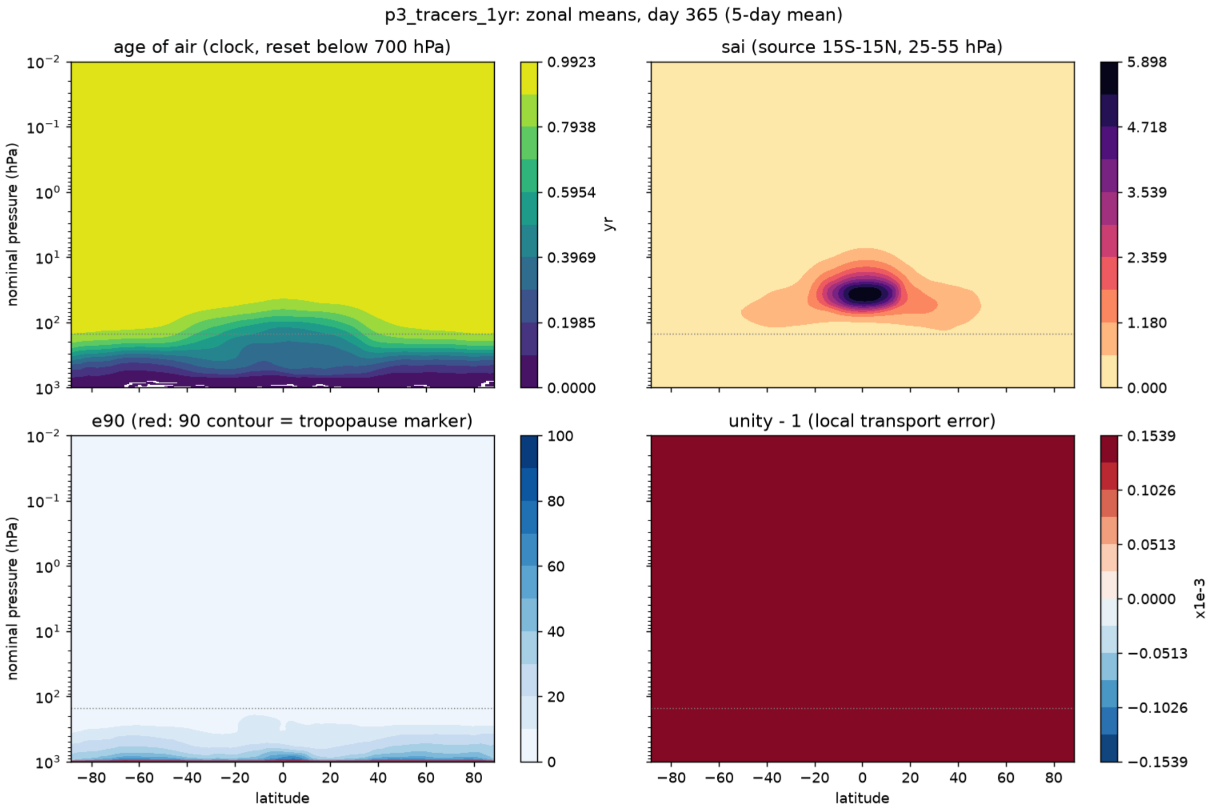

Figure 5. Phase 3 zonal means at day 365. Age of air (top left) reads about one year everywhere above 100 hPa, with only a faint tropical minimum: the stratosphere has aged, fresh air has barely entered. The injection tracer (top right) stays in its source region. e90 (bottom left) decays from the surface instead of marking the tropopause. unity - 1 (bottom right) is uniform to 1.5e-4.

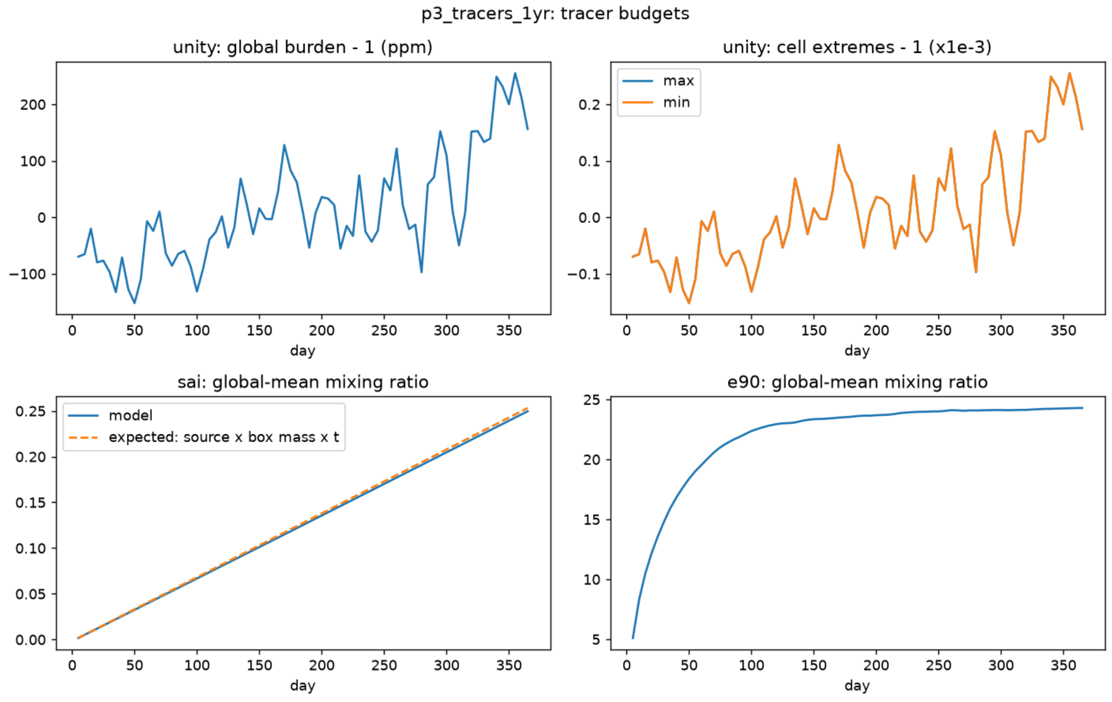

Figure 6. Phase 3 budgets over 2005. unity's global burden and cell extremes stay within 3e-4 of one. The sai burden (bottom left) follows the analytic source line. e90's global mean saturates after about 100 days as expected for a 90-day lifetime.

Phase 4: five years, 2005-2009, and the age of air

Nothing physical changed: the Phase 3 configuration was run from 1 January 2005 to 31 December 2009 (1 826 days). It could not be run as one job. JCM keeps the whole ERA5 nudging target on the GPU, and five years of 6-hourly targets at L95 are 154 GB against an 80 GB card, so the first attempt died in its first step with an out-of-memory error. The five years therefore run as five chained calendar-year segments, each warm-started from the previous segment's checkpoint, which restores the full model state including the tracers. The clock and the injection tracer are continuous to within one save at every year boundary. The segments' output is linked into one 1 826-day run for the analysis. The per-year ERA5 windows were cut locally from one prefetched five-year file rather than downloaded again (issue 26 asks upstream to stream the target).

What passes

- Stability over five years: 65 of 65 chunks healthy, surface pressure drift +0.03 hPa, top-level temperature trend zero.
- Transport: unity within 2.6e-4 of one in every cell at every one of 365 saves and its global burden drifts by 1.7e-4 over five years; the sai burden stays on its analytic line to 1.1 percent; no tracer goes negative beyond roundoff; no polar pull-up (after five years of continuous source the tracer fills the whole upper domain roughly uniformly rather than piling up at the poles).
- Age-of-air pattern: youngest air in the tropics at every level, oldest over both poles; the tropical pipe is there (Figure 7).
- Speed: 4 315 simulated days per hour stepping over the 64 steady chunks (4 460-4 480 within each segment), 1 768 end to end for the whole five years including five compiles and five 31 GB target loads. One simulated year in 10.4 minutes; 30 years in about 5.2 GPU-hours.

What does not

- **The tropics are 1.5 years too old.** At 55 hPa (about 20 km) the model's mean age is 2.9 years between 10S and 10N against 1.3 in CLaMS, while at 50-70 degrees it matches (4.1 vs 4.1). The tropics-to-extratropics contrast is 1.2 years against 2.8 in CLaMS, 43 percent, so the 'within 50 percent' acceptance fails. At 12 hPa the whole profile is 0.2-0.7 years too old (Figure 8). Two causes, both expected from the Phase 1 physics choice. First, weak tropical upwelling: the Held-Suarez stratosphere is isothermal with no meridional temperature gradient and no polar-night jet, so there is little wave-driven Brewer-Dobson circulation above the nudged layer. Second, slow tropospheric transit: the clock resets below 700 hPa but, without convection or boundary-layer mixing, air between 700 and 150 hPa ages for months before it reaches the stratosphere; the tropical profile already reads about one year at 100 hPa where CLaMS reads 0.2. Part of the tropical excess is therefore tropospheric, not stratospheric, and the clock should be referenced to the tropopause for this purpose (issue 25).
- **No polar-night jet, so no SSW test.** The zonal-mean wind at 61N and 10 hPa is -4 to -2 m/s in every winter, where ERA5 has about +30 m/s. The nudged troposphere supplies the wave forcing but the Held-Suarez stratosphere has no vortex for it to disturb, so the three observed warmings of 2006, 2008 and 2009 cannot show up (Figure 9). This is the limitation flagged since Phase 1 (issues 2 and 5); a full-ECHAM specified-dynamics reference year is being run to quantify the gap.
- Five years is still not equilibrium: everything above about 10 hPa reads 4.9-5.0 years, the run age. The upper-stratospheric comparison with CLaMS becomes meaningful only after about 10 years.

What it means

The speed question Approach A was set up to answer is answered: a physics dycore with the troposphere stripped and nudged does 30 years of stratospheric tracer transport in about five GPU-hours on one H100, with the remaining end-to-end cost split roughly half stepping, half output writing (issue 19), and the time-step sweep (issue 3) still untried. An emulator would need to beat a few GPU-hours per 30 years and match this transport fidelity to be worth building. The fidelity bar is currently set by the stratospheric forcing, not by the transport scheme: the same tracers on a stratosphere with a polar-night jet and a realistic Brewer-Dobson circulation are the next experiment.

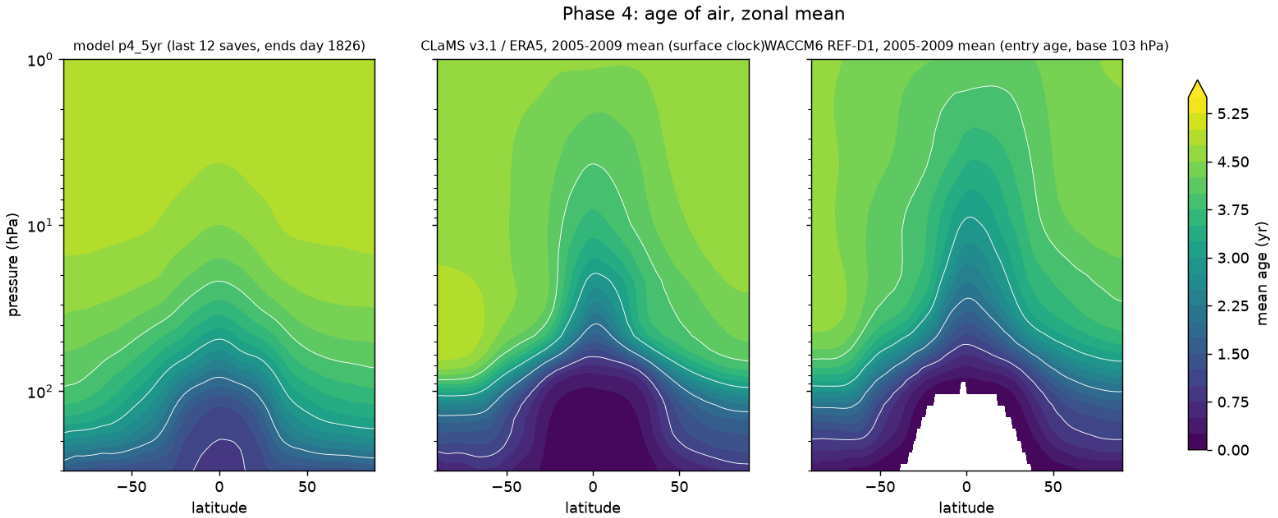

Figure 7. Zonal-mean age of air at the end of 2009: the model (last 12 saves), CLaMS v3.1 driven by ERA5 (2005-2009 mean, a surface clock like ours) and WACCM6 REF-D1 (2005-2009 mean, an entry age relative to 103 hPa, hence younger by construction). The model has the right shape and the right extratropical ages but a tropical pipe that is far too old and too shallow.

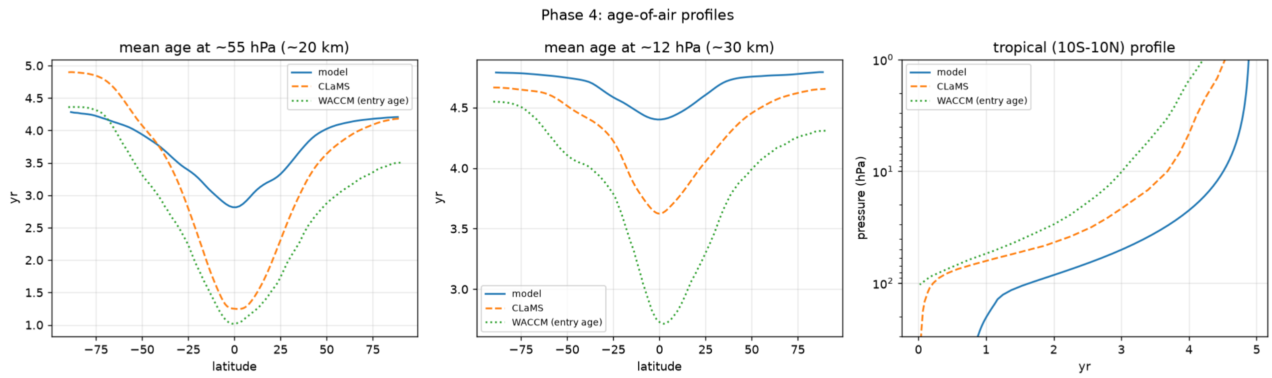

Figure 8. Mean age against latitude at about 55 hPa (left) and 12 hPa (middle), and the tropical vertical profile (right). The extratropics match CLaMS at 55 hPa; the tropics are 1.5 years too old; the tropical profile is already about one year old at 100 hPa, the tropospheric-transit signature.

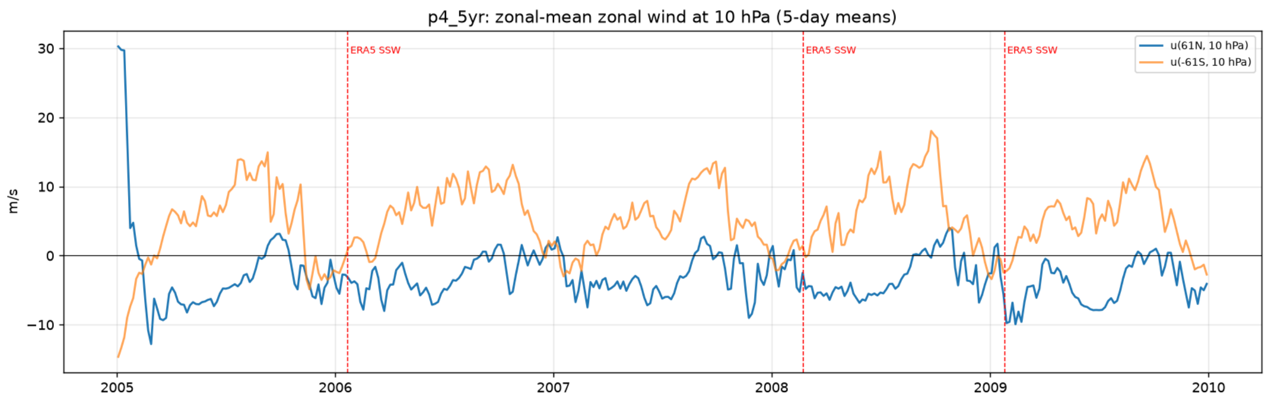

Figure 9. Zonal-mean zonal wind at 10 hPa and 61N (blue) and 61S (orange), 5-day means, with the ERA5 sudden-warming dates marked. There is no winter westerly jet to disrupt in either hemisphere.

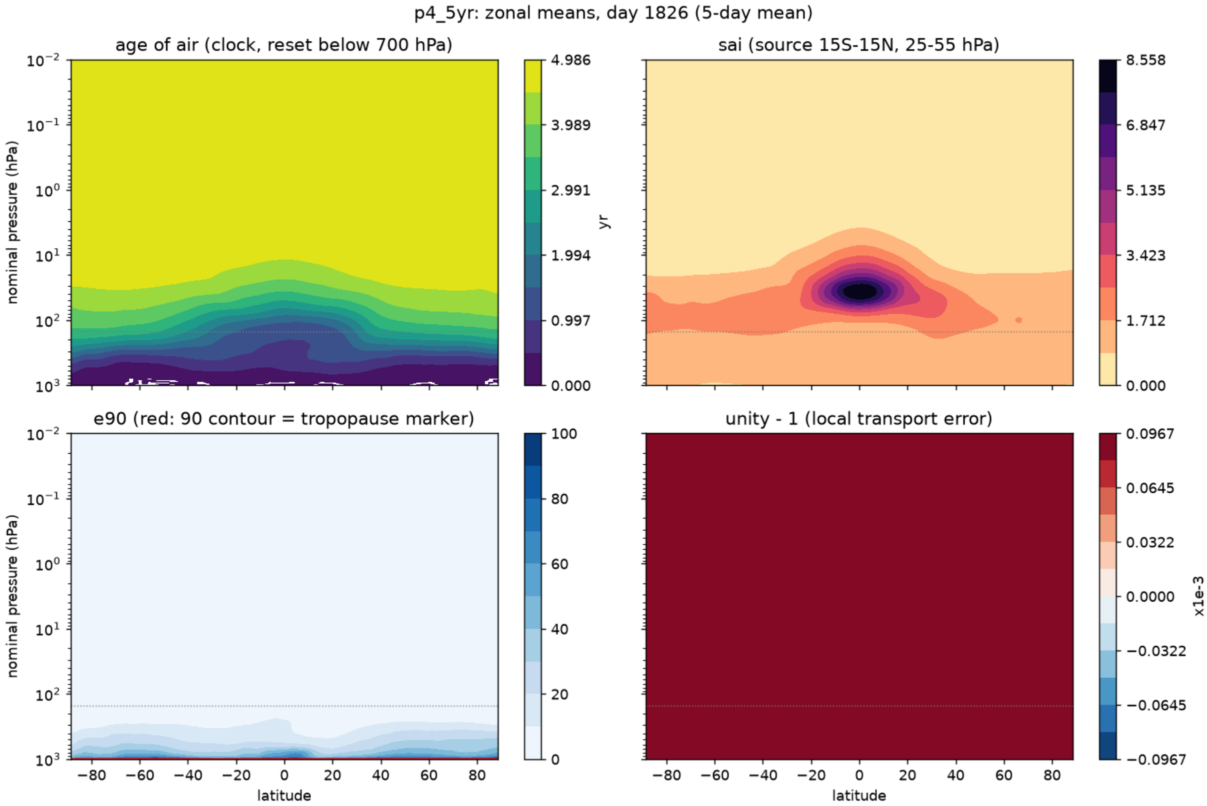

Figure 10. Zonal means at day 1826. Age of air (top left) now shows the tropical pipe through the whole lower stratosphere; the injection tracer (top right) has spread over the entire upper domain after five years of continuous source; e90 (bottom left) still hugs the surface; unity - 1 (bottom right) is uniform to 1e-4.

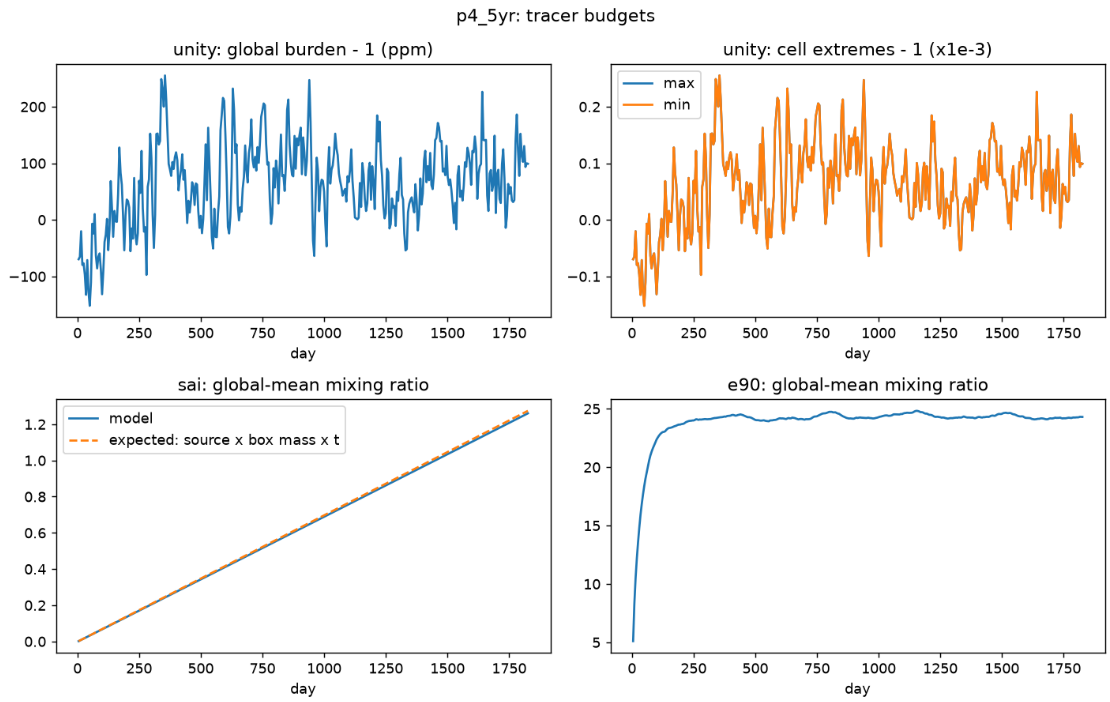

Figure 11. Five-year tracer budgets: unity's burden and cell extremes stay within 3e-4 of one throughout; the sai burden follows the analytic source line for five years; e90 stays saturated.

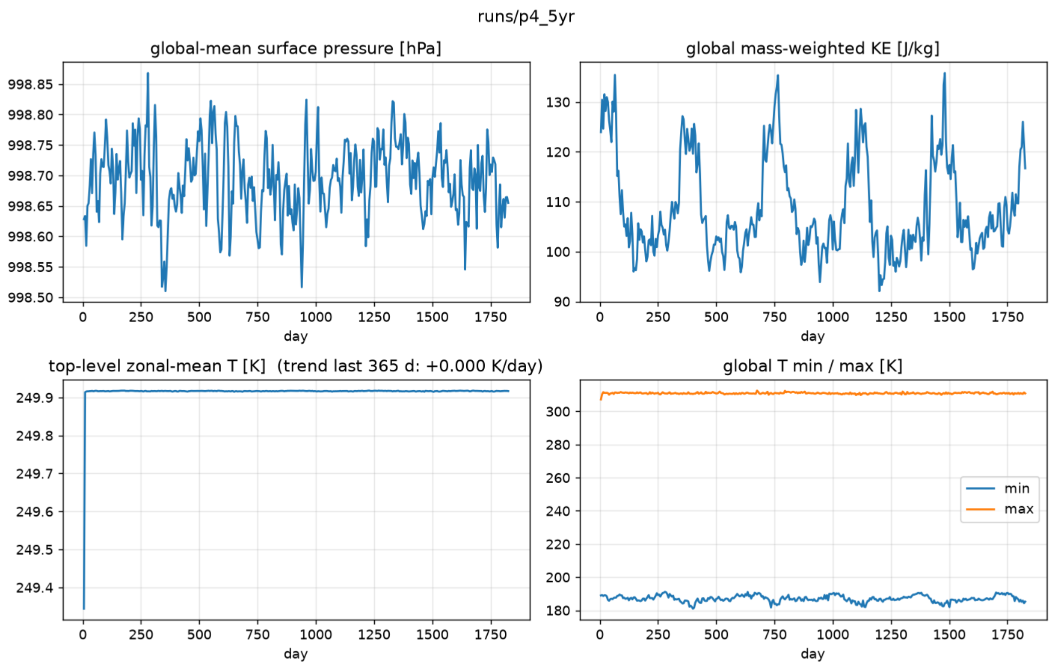

Figure 12. Five-year stability summary: surface pressure, global kinetic energy with its five seasonal cycles, top-level temperature, global temperature extremes.

Speed across the phases

Blue is model stepping only, from per-chunk wall time. Orange is end to end including JIT compile and writing output. The end-to-end number is the one that matters for a 30-year run; the gap between the two is output volume and per-chunk overhead (issue 19). The 12 minute step is far below the semi-Lagrangian ceiling, so a time-step sweep (issue 3) is the next speed lever.

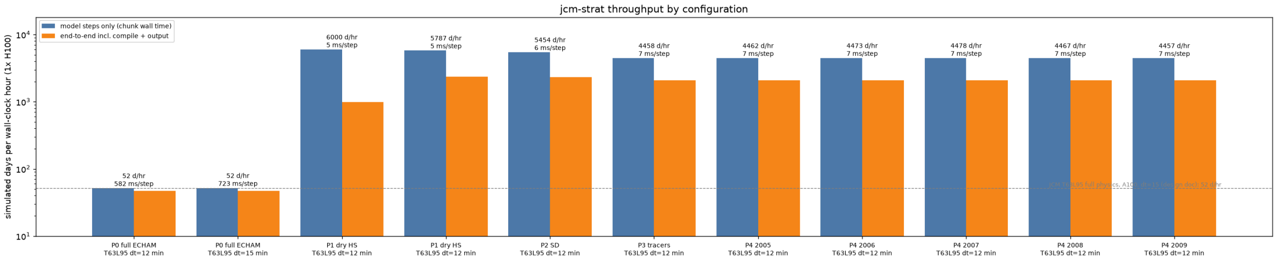

Figure 13. Simulated days per wall-clock hour on one H100 for each configuration, log scale. Removing the physics gave a factor of about 110; nudging cost 6 percent and four tracers a further 18 percent in stepping; the five Phase-4 segments repeat the Phase-3 number to within 1 percent.

Overall reading

The chain is internally consistent: each phase changed one thing and the diagnostics moved only where they should. The transport machinery is cheap, conserving and stable over five years. The one standing scientific weakness is the Held-Suarez stratosphere, isothermal with no polar-night jet and no seasonal cycle, and Phase 4 has now measured its cost: a tropical pipe 1.5 years too old and no vortex. Fixing the stratospheric forcing (Polvani-Kushner, issue 5, or RRTMGP with prescribed gases, issue 2) is the first thing to do next, with the same passive tracers as the yardstick.

Reference data: where it comes from

ERA5 for nudging and initialisation (Phases 2 onward)

Source: the WeatherBench2 public ERA5 archive on Google Cloud Storage, anonymous access, read through JCM's own reader (jcm.data.era5). For the T63 grid (192 longitudes) the reader picks the 240 x 121 equiangular store: `gs://weatherbench2/datasets/era5/1959-2023_01_10-6h-240x121_equiangular_with_poles_conservative.zarr`

- 6-hourly, 1959-2023, 13 pressure levels from 1000 to 50 hPa. Values above 50 hPa are clamped, which is why nudging must stop well below that; the 150 hPa cutoff keeps a wide margin.
- Fields used: u, v, T (nudging target) and the full state for the 1 January 2005 initial condition. Regridded once to the model's hybrid levels and cached under the repository (cache/era5). One year of target is 31 GB and is held on the GPU during the run; five years (154 GB) do not fit an 80 GB card, hence the chained one-year segments in Phase 4.
- Period 2005-2009: volcanically quiet, overlaps the CLaMS record (2004-2023), contains three observed sudden stratospheric warmings (January 2006, February 2008, January 2009).

ERA5 zonal-mean tape for the stratospheric comparison rows

The bottom rows of Figures 2 and 4 compare against a monthly zonal-mean ERA5 tape from the Copernicus Climate Data Store product reanalysis-era5-pressure-levels-monthly-means, prepared earlier for the AIDE atmosphere validation. It holds zonal-mean U and T for 1989-1994 on six levels between 7 and 70 hPa at 0.25 degree latitude. Location: `/home/susanne/docs/AIDE-atmosphere_validation/AIDE-atmosphere/output/era5_monthly_tape.nc`

Note that the tape years (1989-1994) differ from the model years (2005-2009). Since the Held-Suarez stratosphere has no interannual variability this is a bias check, not a year-matched comparison.

CLaMS age of air (Phase 4 reference)

Chemical Lagrangian Model of the Stratosphere, version 3.1 (Konopka et al. 2025), driven by ERA5, from the Zenodo dataset 'Gridded CLaMS Simulations Driven by Multiple Reanalyses' (doi:10.5281/zenodo.17357000, PI F. Ploeger, Forschungszentrum Juelich). Monthly zonal means on 39 pressure levels and a 1 degree latitude grid, 2004-2023. The variable AGE is mean age from a clock tracer increasing linearly at the Earth's surface, in years. Local copy: `/data/CLaMS/CLaMS_v3/clams_v3.1_era5_zm_lat.zip` (one file per year).

Note the definitional difference from our clock: CLaMS resets at the surface, jcm-strat resets everywhere below 700 hPa, and WACCM (below) measures age relative to an entry point at about 100 hPa. CLaMS is the like-for-like reference; WACCM is shown for the pattern. Referencing our clock to the tropopause is issue 25.

WACCM6 age of air (Phase 4 reference)

CESM2 WACCM6 REF-D1 simulations (the CCMI-2022 reference historical experiment), ensemble member 04, AOA1mf clock tracer, 1970-2019, 70 levels, 192 latitudes, noleap calendar. Age is given in years relative to a base point at latitude 0.47 degrees, 103 hPa. Members 01-03 and SF6-derived ages are also available in the SAVE subdirectory. Local copy: `/data/CESM2_REFD1_AOA/AoA_waccm6_refd1.04_AOA1mf_1970-2019_ba_0_100.0_ck_0_50.0.nc`

Model code and pins

- JCM (climate-analytics-lab/jax-gcm), branch dev, commit 849893b, 1 September 2026, as a submodule.
- Dinosaur (shoyer/dinosaur), branch semi-lagrangian, commit bd99e39b, 24 August 2026, as a submodule; the open upstream PR is neuralgcm/dinosaur#135.
- Every phase's exact command, resolved configuration, commit and wall-clock time is in docs/outputs/<phase>/output.md in the repository, with the plots shown here beside it. This PDF is generated from those records by scripts/make_report.py.